

CTRL-C2000 control card

Rev E0 | 2026-10-04 | Schematics authored by Chinmoy Bhuyan | TMS320F28388D common controller for PV-P75/100/110 and DAB-D60

01 Power regulators

+24V -> LMR38020 5 V -> TPS62133 3.3 V; PG chain enables TPS62130 1.2 V and TPS7A2033 3.3 V analog (VDDIO first, then VDD and VDDA)

File: sheets/01_power.kicad_sch

02 Supervisors, reset, clock, boot

TPS3703 windows on 3.3 V and 3.3 V analog, UV supervisor on 1.2 V -> XRSn; 25 MHz CMOS clock into X1; boot straps GPIO72/GPIO84 (flash by default)

File: sheets/02_reset.kicad_sch

03 TMS320F28388D (ZWT 337-ball)

All 337 balls, pin table extracted from SPRSP14E (gen/data/f28388d_pins.csv); net per ball from the pin plan (gen/data/ctrl_c2000_pin_plan.csv)

File: sheets/03_mcu.kicad_sch

04 MCU supply decoupling

SPRSP14E 7.10.1.5: 0.1 uF per VDDIO/VDD3VFL/VDDOSC ball, 2.2 uF per VDDA ball, -22 uF total on VDD + one 56 R load, 22 uF per VREFHI (16-bit mode)

File: sheets/04_decoupling.kicad_sch

05 ADC reference, VMID, slow mux

REF5025E 2.5 V for VREFHIA-D; OPA2388 buffers VMID 1.25 V (diff-amp offset) and VBIAS 2.5 V (ID/NTC bias); TMUX1208 for 4 NTC + 4 ID; 24 V monitor; AGND star

File: sheets/05_adcref.kicad_sch

06 Fault receivers, enable, EN drivers

Cn_EN = GATE_EN AND CHB_OK AND MCU enable (74LVC11A); line drivers on only while CHB_OK, MCU enable and RDY; EN drivers with readback; fail-safe FLT receivers

File: sheets/06_trip.kicad_sch

07 Cell port C1

CELL contract 2x20: 8 PWM RS-422 drivers, 3 dual-range analog inputs, RDY, ID, NTC; gate-driver supply +24V_GD through a 1.5 A eFuse (TPS26600)

File: sheets/07_cell1.kicad_sch

08 Cell port C2

CELL contract 2x20: 8 PWM RS-422 drivers, 3 dual-range analog inputs, RDY, ID, NTC; gate-driver supply +24V_GD through a 1.5 A eFuse (TPS26600)

File: sheets/08_cell2.kicad_sch

09 Cell port C3

CELL contract 2x20: 8 PWM RS-422 drivers, 3 dual-range analog inputs, RDY, ID, NTC; gate-driver supply +24V_GD through a 1.5 A eFuse (TPS26600)

File: sheets/09_cell3.kicad_sch

10 Cell port C4

CELL contract 2x20: 8 PWM RS-422 drivers, 3 dual-range analog inputs, RDY, ID, NTC; gate-driver supply +24V_GD through a 1.5 A eFuse (TPS26600)

File: sheets/10_cell4.kicad_sch

11 Port-board analog inputs

PORT contract 2x13: VA, VAX, VB, VBX, IA, IB, AUX1, AUX2 isolated-amplifier pairs, biased open-safe, through difference amp + RC + clamp; ID: +24V out through an eFuse

File: sheets/11_port.kicad_sch

12 SYS-I0-AUX connector

80-way SYS contract: +24 V logic, +24V_GD (channel B), CAN, SCI, SPI, I2C, DI/DO, fault latch, GATE_EN, CHB_OK, watchdog, reset, fans, Ethernet MDI + centre tap

File: sheets/12_sys.kicad_sch

13 Ethernet PHY DP83822I

MII to the CM EMAC (GPIO105-124, mux 14), own 25 MHz oscillator, 49.9 R MDI terminations to AVD; MDI pairs, LEDs and the centre-tap supply ETH_CT go to SYS

File: sheets/13_eth.kicad_sch

14 JTAG, UART, EEPROM, LEDs, SDFM

cTI-20 JTAG, CM debug UART, 24LC256 calibration EEPROM (0x57), status LEDs, ERRORSTS, 8-channel SDFM expansion header (ECO-02 / ECO-09)

File: sheets/14_debug.kicad_sch

Protection without firmware: FLT_LATCH_N -> INPUTXBAR1 -> TZ1; each cell's FAULT (INPUTXBAR2-5) and AN1 CMPSS window -> TRIP4/5/7/8; port OV (ADC PPB on VA/VB, CMPSS backup), open inputs (CMPSS low), IA/IB SC (CMPSS) and OC (PPB), P_ID (PPB) -> TRIP9 for every cell; E-stop -> INPUTXBAR6 -> TRIP6. CMPSS reference = VDAC = REF5025E; firmware nulls each channel at zero input at start-up; trip bands in the pin plan, asserted against the frozen earlier-platform snapshot gen/data/control_spec_platform1.json (D-044). EN pair = GATE_EN AND CHB_OK AND MCU enable (74LVC11A); line drivers on only with CHB_OK, MCU enable and RDY; EN readback on GPIO100-103. +24V_GD feeds only the cell eFuses.

Rails: 24 V -> 5 V (LMR38020) -> 3.3 V (TPS62133); its PG enables 1.2 V (TPS62130) and 3.3 V analog (TPS7A2033), so VDDIO leads VDD and VDDA (SPRSP14E table 7-1). Three TPS3703 hold XRSn (windows on 3.3 V and 3.3 V analog, UV on 1.2 V), every tolerance corner asserted. REF5025E runs from the analog rail, so VREFHI and VDAC never exceed VDDA.

Pin plan with mux positions and trip bands: gen/data/ctrl_c2000_pin_plan.csv (checked against SPRSP14E table 6-1 on every build). Pin table: gen/data/f28388d_pins.csv (from the PDF).

SYS side: inputs via 74LVC244A / 74LVC1G17 (loff); I2C pull-ups and EEPROM 0x57 here; ETH_CT = PHY AVD; PORT +24V via its own eFuse.

Not bench-validated. Values marked calculated are not simulated.
Sheet 1
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

24 V -> 5 V (LMR38020, 400 KHZ)

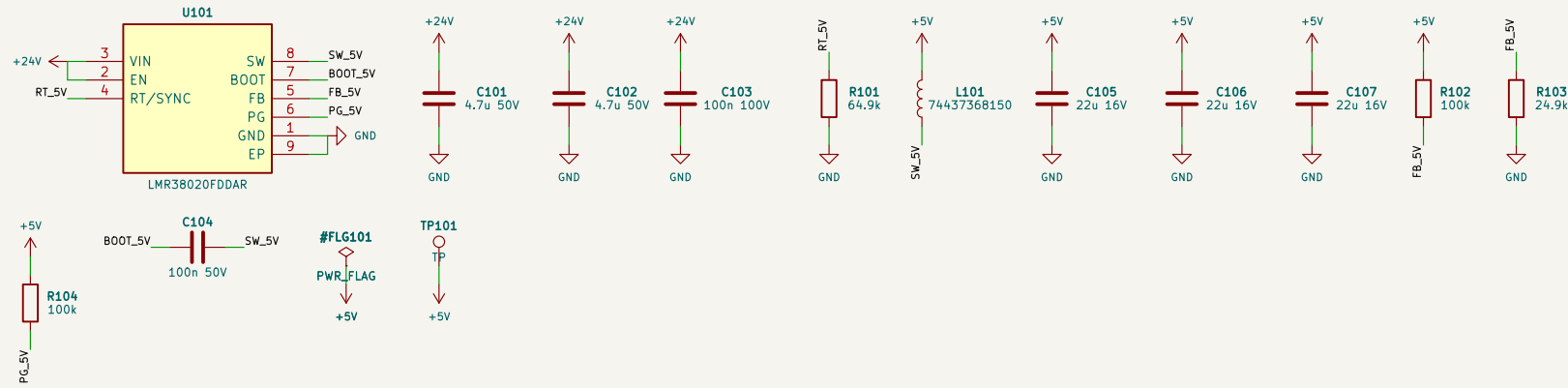
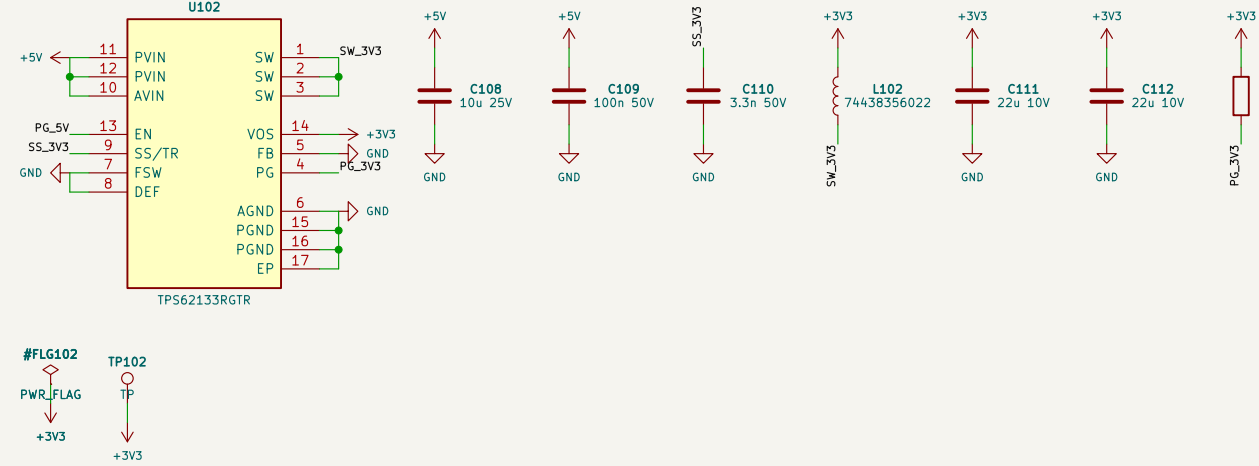


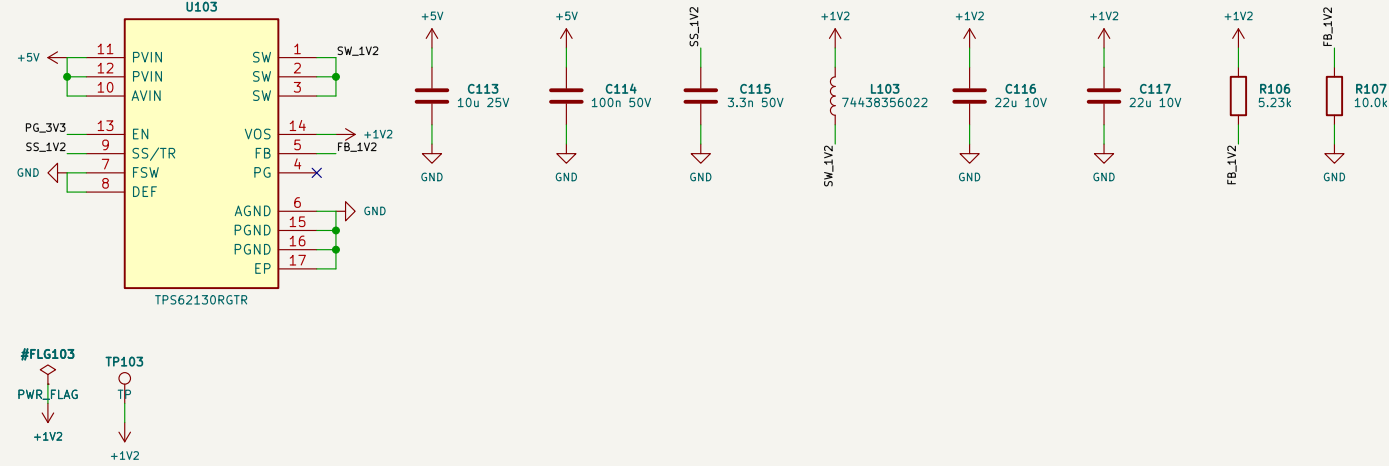
Table 9-1 row 400 kHz / 5 V / 2 A: L 15 uH, 3x22 uF, RFBT 100k / RFBB 24.9k -> 5.02 V, RT 64.9k.
Load (calculated): 3.3 V 1.3 A + 1.2 V 0.48 A + 3.3 V analog 0.1 A -> about 1.2 A at 5 V.
EN tied to VIN (datasheet allows), PG_5V enables the 3.3 V converter.
Output capacitance check (SNVSC40E, calculated): total on +5V = 3x22 + 2x10 + 1 + 0.3 = 87 uF nominal,
96 uF at +10 %; 1.45x the 66 uF design value, inside the 9.2,2.5 limit (10x design or 1000 uF) -> no extra
stability study required. Start-up: tSS 4.0 ms (8.3.9) -> 96 uF x 5.02 V / 4 ms = 0.12 A charge current,
downstream loads held off by the PG chain, << ILS-LIMIT 1.8 A min; OCP blanking 18 ms -> no hiccup.
Inductor: LMIN = 0.25 x 5 V / 400 kHz = 3.1 uH (eq. 11) < 15 uH; ripple 0.64-0.68 A at 21.6-26.4 V in
(> 10 % of 2 A); peak 1.54 A (1.62 A at L -20 %) < IHS-LIMIT 2.6 A min < ISAT 5.55 A. Bode not measured.

5 V -> 3.3 V DIGITAL (TPS62133, 3 A)



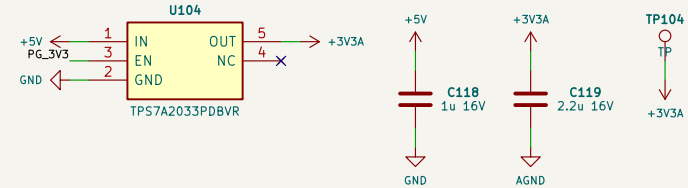
Fixed 3.3 V; FB to AGND per datasheet, FSW low (2.5 MHz), 2.2 uH, 2x22 uF (table 9-1).
SS 3.3 nF (typical application), PG_3V3 releases the 1.2 V and 3.3 V analog rails.
Worst case 3.221-3.396 V (power-save mode, SLVSAG7F p5) inside VDDIO 3.14-3.47 V.

5 V -> 1.2 V CORE (TPS62130, 3 A)



VOUT = 0.8 V x (1 + 5.23k/10.0k) = 1.218 V (eq. 6), 0.1 % 10 ppm/K; worst case 1.1872-1.2558 V
(power-save VFB 781.6-822.4 mV, FB leakage, 0.1 % line/load) inside VDD 1.14-1.26 V (CSR-01).
Enabled by PG_3V3: VDD rises after VDDIO (SPRSP14E table 7-1). VDD bulk on the decoupling sheet.

5 V -> 3.3 V ANALOG (TPS7A2033)



VDDA, op amps, reference, mux. Enabled by PG_3V3 so VDDA never leads VDDIO.
Load (calculated): 26 op-amp channels x 2.6 mA max + IDDA 15 mA + REF and its 4.2 mA load + VBIAS
+ 12 open-input bias resistors to +3V3A (16 mA) = about 110 mA of 300 mA.

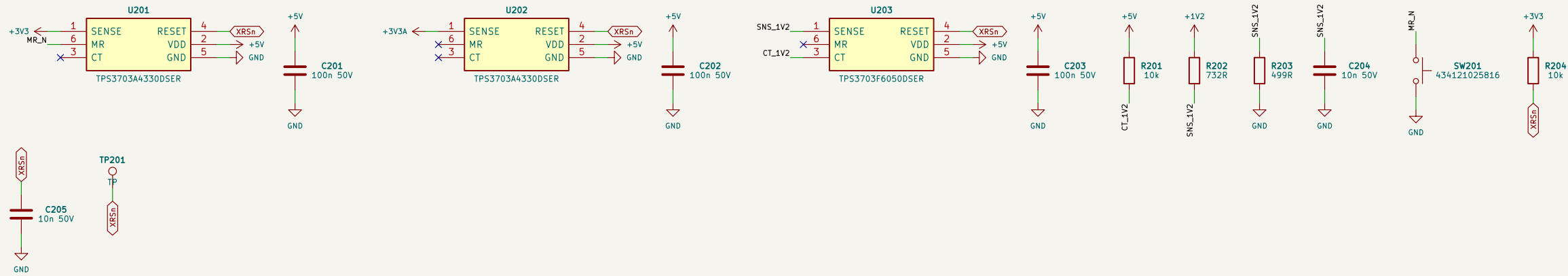
Not bench-validated. Values marked calculated are not simulated.
Sheet 2
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

Sheet: /01 Power regulators/
File: 01_power.kicad_sch

Title: CTRL-C2000 control card - Power regulators

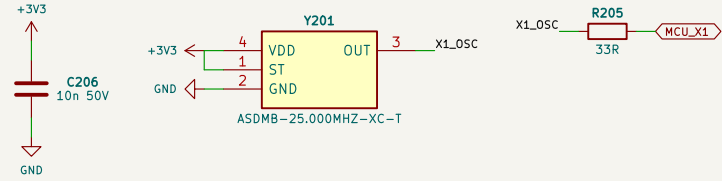
Size: A3 Date: 2026-10-04 Rev: E0
KiCad E.D.A. 10.0.6 Id: 2/15

SUPPLY SUPERVISORS (OPEN-DRAIN RESETS WIRED TO XRSN), CSR-01



SPRSP14E 7.10.1.2.4: VDDIO and VDD need an external supervisor (the internal POR trips below range). Worst case at every corner (asserted on each build, MCU window 3.14–3.47 V / 1.14–1.26 V): +3V3 A4330: trips ≥ 3.1458 , releases $\leq 3.2157 < \text{regulator min } 3.2209$; OV 3.408–3.456 $> \text{regulator max } 3.3958$ V +3V3A A4330: trips ≥ 3.1458 , releases $\leq 3.2157 < \text{regulator min } 3.2505$; OV 3.408–3.456 $> \text{regulator max } 3.3495$ V +1V2 F6050: trips ≥ 3.1458 , releases $\leq 3.2157 < \text{regulator min } 1.1872$; regulator max 1.2558 V 1.2 V is UV-only: a $\pm 4\%$ window cannot hold the TPS62130 power-save spread; its maximum is the regulator arithmetic above. Delay: CT open 10 ms (A parts), CT to VDD via 10k 20 ms (F part). VDD = +5V so the outputs stay defined while 3.3 V collapses. XRSn pull-up 10k, 10 nF; SYS_RST_N and the JTAG RESET are open drain on the same node. Reset button on MR of the 3.3 V part (100k internal pull-up).

25 MHZ CLOCK (X1)



SPRSP14E 7.10.3.2.1: X1 from an oscillator 10–25 MHz, tr/ff ≤ 6 ns, 45–55 %; X2 open (figure 7–11). No crystal, so VSSOSC goes to board ground (table 6–12). 33 R source termination.

BOOT MODE (GPIO72 = PIN 1, GPIO84 = PIN 0)

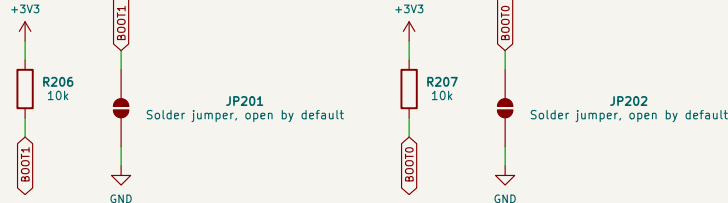
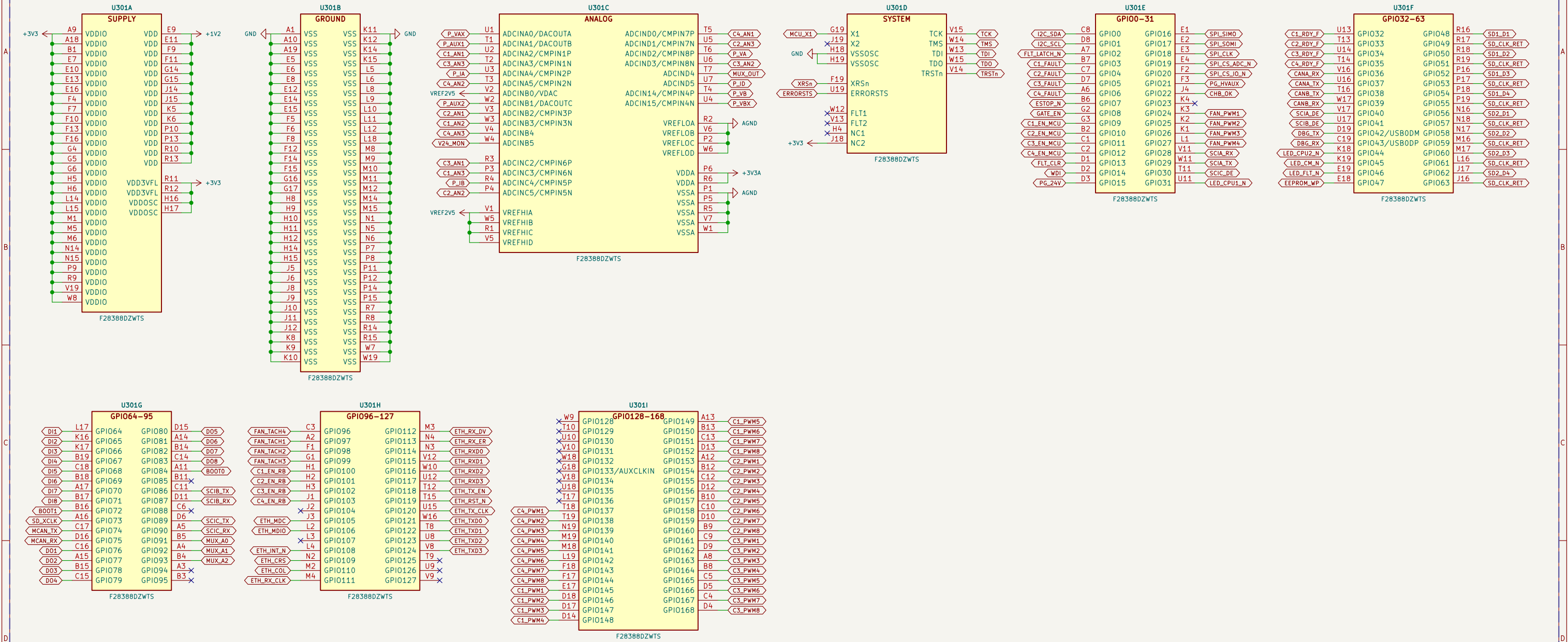


Table 8–18: 11 = flash (straps open), 01 = SCI, 10 = CAN, 00 = parallel I/O. Close JP to pull a pin low. th(boot-mode) 1.5 ms; nothing else drives these balls.

Not bench-validated. Values marked calculated are not simulated.
Sheet 3
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

U301 TMS320F28388DZWTs



ZWT, not PTP-176: all 32 ePWM outputs sit on ZWT-only balls GPIO137-168 (PTP has them only on GPIO0-31, colliding with SPI/SDFM/CAN/eQEP), and ZWT bonds all 24 ADC inputs (PTP lacks B4, B5, C5, D5). HRPWM exists on ePWM1-8 only = ports C1 and C2. Unused GPIOs: no-connect (table 6-12, firmware pull-up). NC2 (J18) tied to VDDIO as the datasheet recommends; FLT1/FLT2 must stay open.

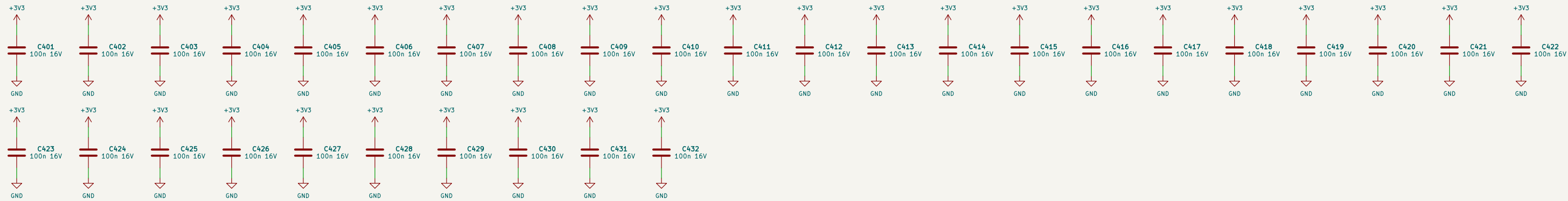
Not bench-validated. Values marked calculated are not simulated.
Sheet 4
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

Sheet: /03 TMS320F28388D (ZWT 337-ball)/
File: 03_mcu.kicad_sch

Title: CTRL-C2000 control card - TMS320F28388D (ZWT 337-ball)

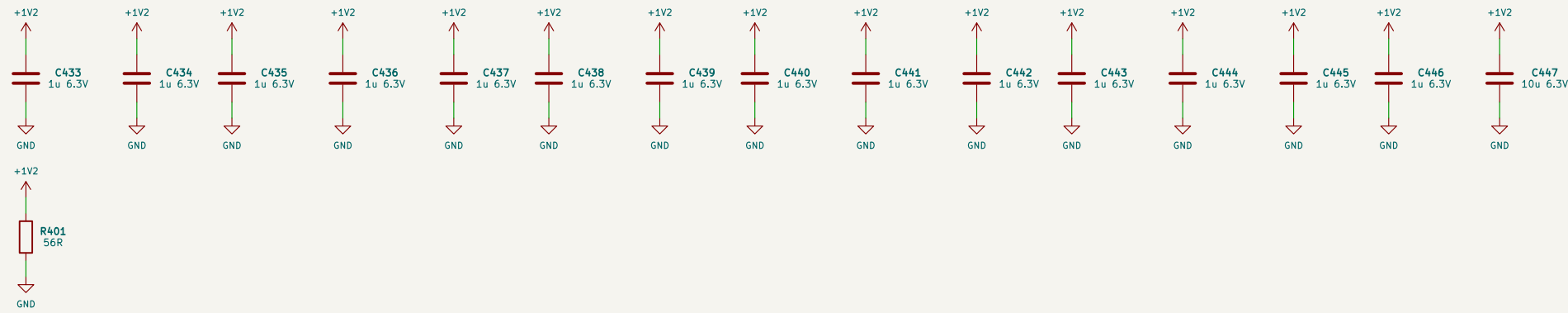
Size: A3 Date: 2026-10-04 Rev: E0
KiCad E.D.A. 10.0.6 Id: 4/15

VDDIO (28 BALLS) + VDD3VFL (2) + VDDOSC (2): 0.1 uF EACH



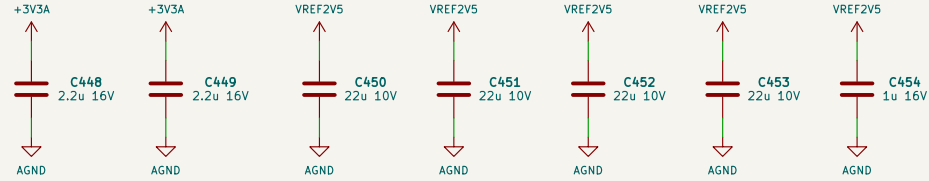
Configuration 1 of 7:10.1.3.2.

VDD 1.2 V (14 BALLS)



14 x 1 uF + 10 uF = 24 uF, 10 % parts: >= 21.6 uF (CVDD TOTAL 20 min / 22 typ. max tolerance 20 %).
56 R load consumes the VDD3VFL-to-VDD leakage current (table 6-4 note).

VDDA, VREFHI AND VDAC



VDDA 2.2 uF per ball to VSSA. VREFHIA-D: 22 uF each to VREFLO (16-bit mode).
VDAC (ADCINB0, V2) = CMPSS DAC reference: >= 1 uF at the ball (table 6-1).
All on VREF2V5: 4 x 22 + 1 + 1 uF = 90 uF, 99 uF at +10 %: inside the REF5025E 1-100 uF range (summed from the design on every build).

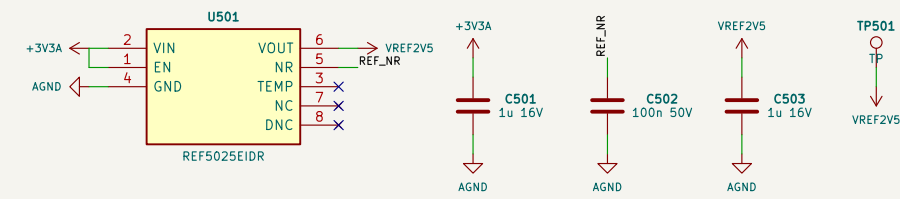
Not bench-validated. Values marked calculated are not simulated.
Sheet 5
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

Sheet: /04 MCU supply decoupling/
File: 04_decoupling.kicad_sch

Title: CTRL-C2000 control card - MCU supply decoupling

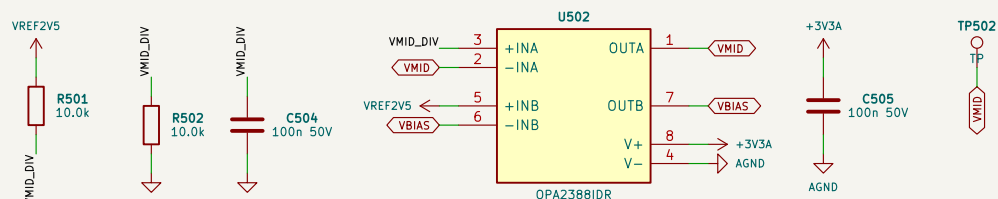
Size: A3	Date: 2026-10-04	Rev: E0
KiCad E.D.A. 10.0.6		Id: 5/15

2.5 V ADC AND CMPSS DAC REFERENCE



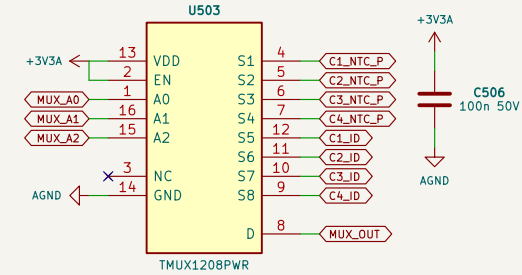
Powered from +3V3A (VIN >= VOUT + 0.2 V) so VREFHI and VDAC can never exceed VDDA (SPRSP14E 7.11.2.3.5, 7.11.3.1.4). Also the CMPSS DAC reference (VDAC, INT-05); thresholds ratiometric with the ADC.
Load 4.2 mA of +/-10 mA: VDAC 8 x 6 k (3.3 mA) + VREFHI 4 x 190 uA + VMID divider (asserted).
NR 0.1 uF class 1; 1 uF HF at VOUT; VREFHI and VDAC capacitors are on the decoupling sheet.

VMID AND VBIAS BUFFERS



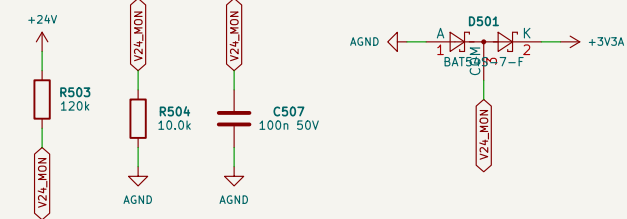
VMID = VREF/2 is ratiometric: a zero differential input reads mid-scale whatever the reference error.
VBIAS = VREF buffered, so ID and NTC readings are ratiometric and the reference itself is not loaded.

SLOW-SIGNAL MUX -> ADCIND4



S1-S4 = C1-C4 NTC, S5-S8 = C1-C4 ID (24 ADC balls cannot take 20 fast channels + 9 slow ones).
Each input has its own 100 nF on the cell sheet; the mux only switches between charged caps.

24 V MONITOR -> ADCINB5



120k/10k: 24 V reads 1.85 V, full scale 32.5 V: 100 nF filter.
+24V is live before VDDA: the BAT54S holds the ball at VDDA + VF (7.10.1.4.2).

AGND - GND STAR POINT



Single connection of the analog return to board ground (layout: under U301).

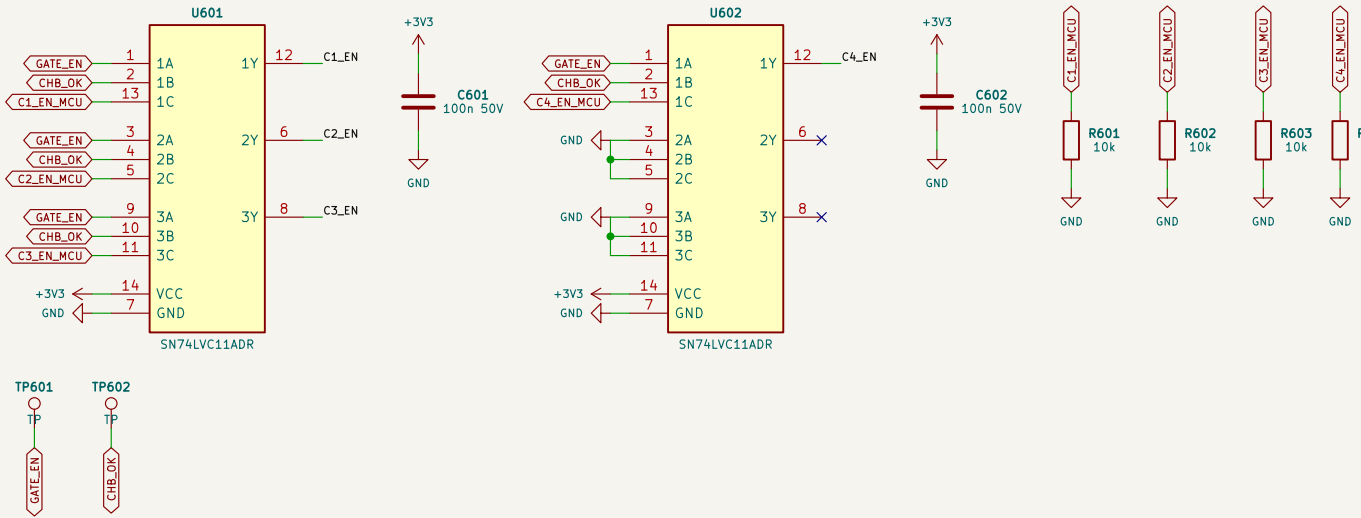
Not bench-validated. Values marked calculated are not simulated.
Sheet 6
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

Sheet: /05 ADC reference, VMID, slow mux/
File: 05_adcref.kicad_sch

Title: CTRL-C2000 control card - ADC reference, VMID, slow mux

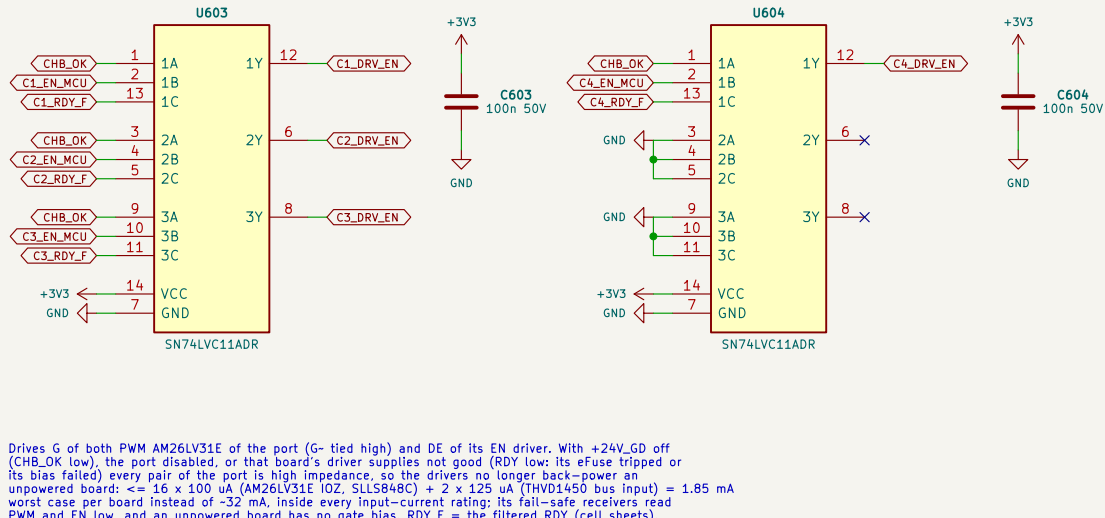
Size: A3 Date: 2026-10-04 Rev: E0
KiCad E.D.A. 10.0.6 Id: 6/15

HARDWARE GATE-ENABLE AND (BOTH SAFETY CHANNELS)



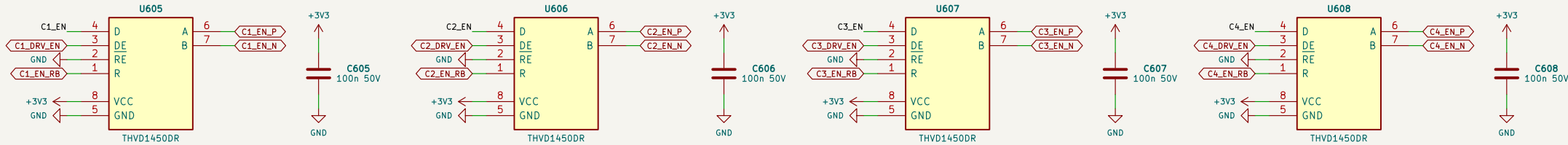
Cn_EN = GATE_EN (channel A) AND CHB_OK (channel B) AND Cn_EN_MCU, one 3-input gate per port; firmware alone can never raise an EN pair. Channel B also removes +24V_GD (gate-driver bias) on SYS. 10k pull-downs hold each MCU enable low in reset; GATE_EN and CHB_OK are pulled low on the SYS side. Spare gates: inputs to GND, outputs open.

LINE-DRIVER ENABLE CN_DRV_EN = CHB_OK AND CN_EN_MCU AND CN_RDY (INT-15)



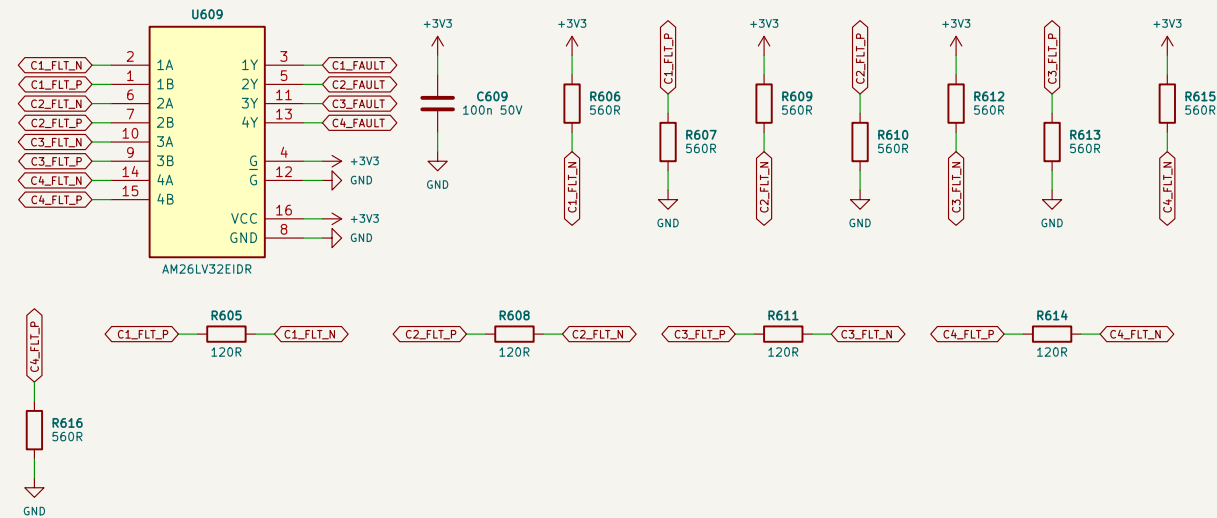
Drives G of both PWM AM26LV31E of the port (G- tied high) and DE of its EN driver. With +24V_GD off (CHB_OK low), the port disabled; or that board's driver supplies not good (RDY low: its eFuse tripped or its bias failed) every pair of the port is high impedance, so the drivers no longer back-power an unpowered board: $\leq 16 \times 100 \mu A$ (AM26LV31E IOZ, SLLS848C) + $2 \times 125 \mu A$ (THVD1450 bus input) = 1.85 mA worst case per board instead of -32 mA, inside every input-current rating: its fail-safe receivers read PWM and EN low, and an unpowered board has no gate bias. RDY_F = the filtered RDY (cell sheets).

EN LINE DRIVERS WITH READBACK (C1-C4)



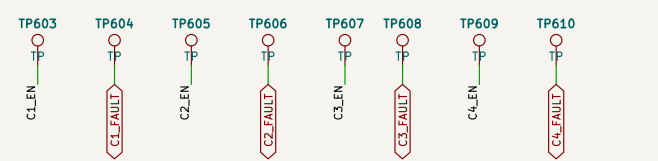
THVD1450 per port: D = Cn_EN, DE = Cn_DRV_EN, A/B = EN_P/EN_N, [VOD] ≥ 2 V into 100 R (SLLSEY3E 7.7). Receiver always on (RE- low): R = the pair's state at the CTRL end $\rightarrow$ Cn_EN_RB on a spare GPIO, so a stuck AND gate or driver is visible (INT-22: start-up and proof-test sequence in the pin plan).

FLT RECEIVERS (C1-C4), FAIL-SAFE



Pair wired swapped (A = FLT_N, B = FLT_P): Cn_FAULT is high = fault. 560R bias + 120R termination put +0.32 V across an open/unplugged pair ($\rightarrow$ VIT+ 0.2 V) $\rightarrow$ fault; the AM26LV32E rev E only has open-circuit (not terminated) fail-safe, hence the external bias. Healthy: driver -2 V $\rightarrow$ low, Cn_FAULT $\rightarrow$ INPUTXBAR2-5 $\rightarrow$ ePWM X-BAR $\rightarrow$ TRIP4/5/7/8 (no firmware in the trip path).

TEST POINTS



Gated EN (after the AND) and receiver FAULT outputs, one pair per port.

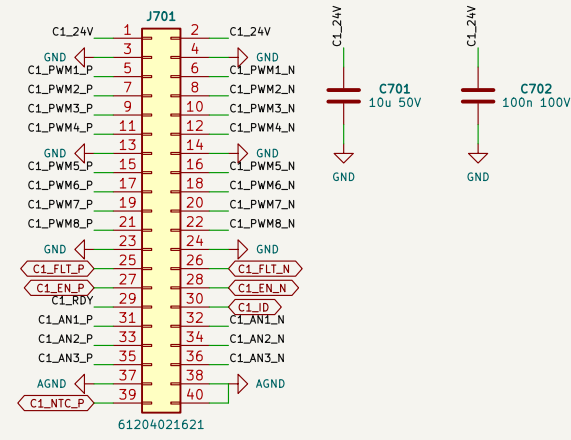
Not bench-validated. Values marked calculated are not simulated.
Sheet 7
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

Sheet: /06 Fault receivers, enable, EN drivers/
File: 06_trip.kicad_sch

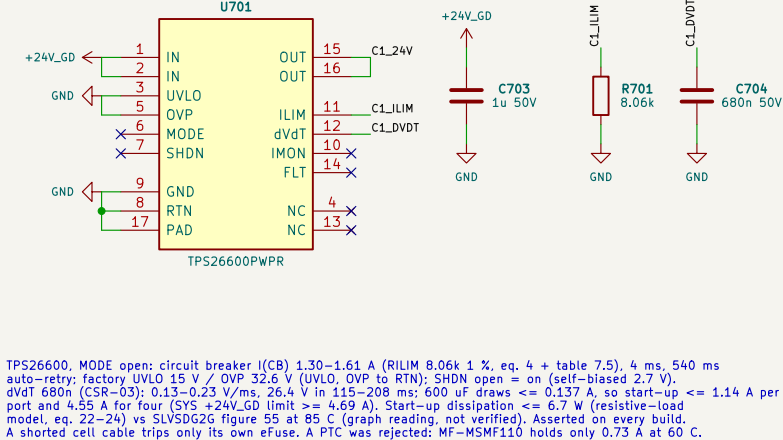
Title: CTRL-C2000 control card – Fault receivers, enable, EN drivers

Size: A3 Date: 2026-10-04 Rev: E0
KiCad E.D.A. 10.0.6 Id: 7/15

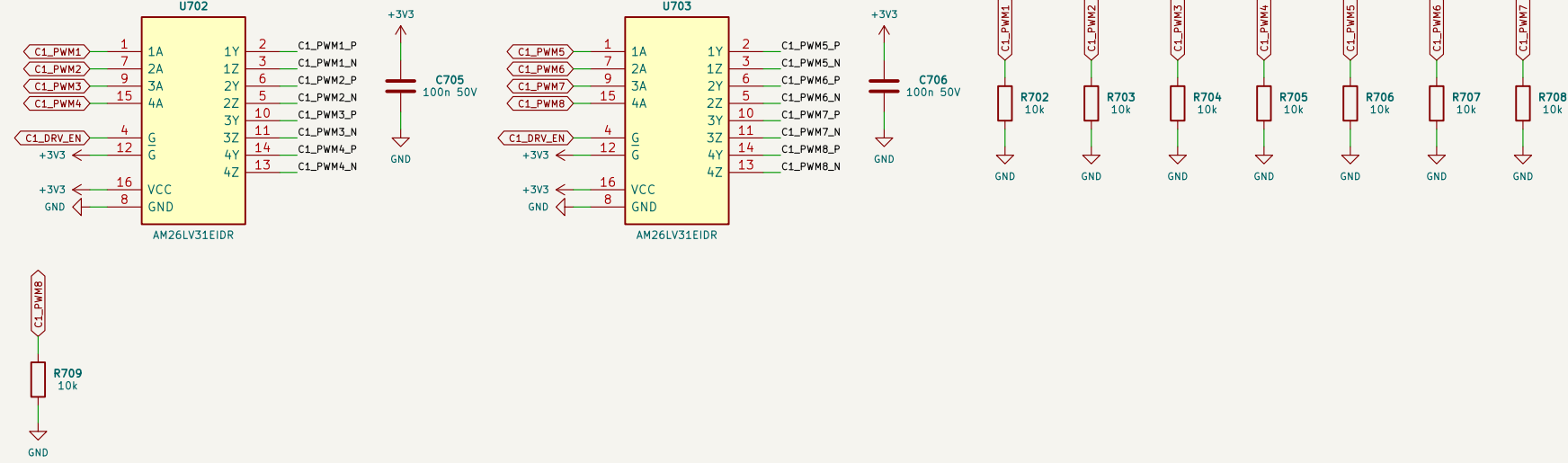
C1 CONNECTOR (CELL CONTRACT)



+24V_GD EFUSE FOR C1 (<= 1.0 A CONTINUOUS, <= 600 uF)

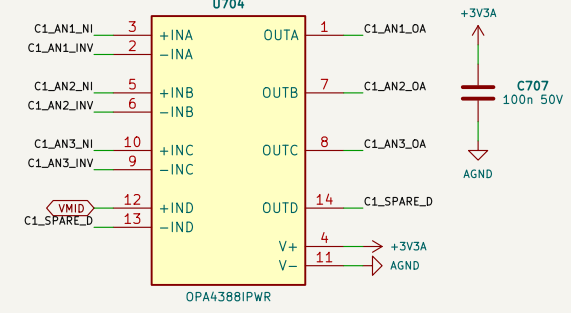


PWM LINE DRIVERS (EPWM1-4)



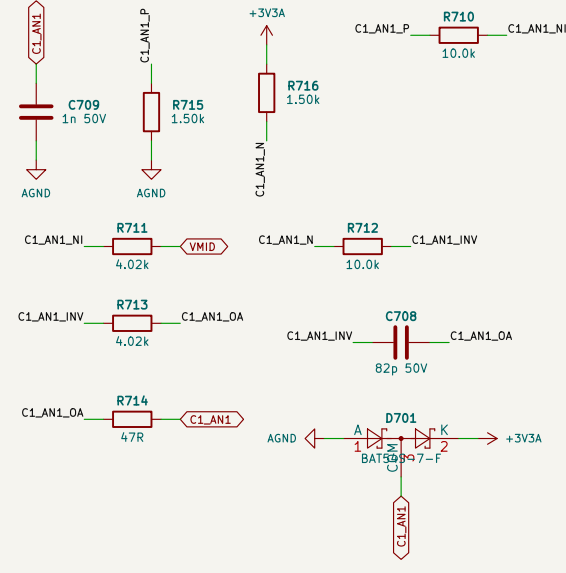
PWMx (1 = switch on) -> RS-422 pair, terminated on the power board. 10k pull-downs; ePWM balls are inputs without pull-up during reset/boot (table 6-6), so every gate command is 0 until firmware runs. G = CL_DRV_EN, G= high: outputs high impedance unless CHB_0K, CL_EN_MCU and RDY are high (N1-15). ePWM1-4 have high-resolution PWM: the dual-active bridge (phase shift at 100 kHz) must use ports C1 and C2, one bridge per port.

ANALOG FRONT ENDS AN1-AN3 (OPA4388), DUAL RANGE

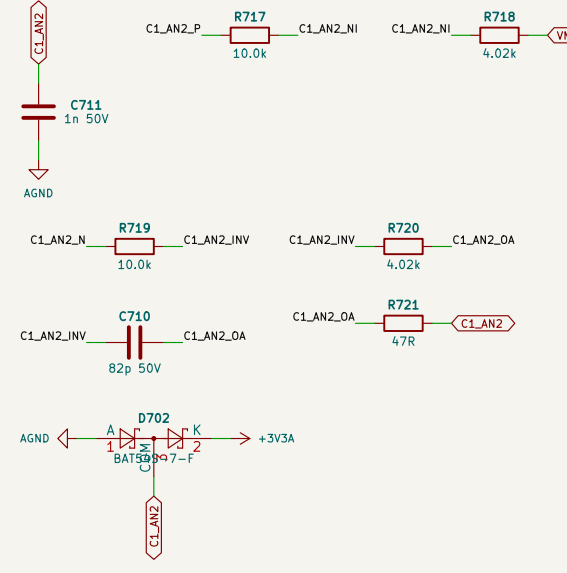


Difference amp per pair: $V_{ad} = V_{MD} 1.25 V + G \times (ANL_P - ANL_N)$, Rin 10.0k 0.1 % 10 ppm/K, $G = 4.02k/10.0k = 0.402$. C1 || Rf anti-alias pole 483 kHz, 47 R + 1 nF CG; ADC charge bucket (ACQPS >= 320 ns at 12 bit). BAT5MS clamps the ADC node to 3V3A/AGND. All values calculated, not simulated. Accepts on any AN pair: differential on a 1.24-1.49 V common mode, legs 0.1-2.6 V (ADC span +/-3.1 V), or 0-3.0 V single-ended with AN_N = AGND at the source (temperatures) -> 1.25-2.46 V, Op-amp inputs 0.9-1.75 V (RMID). Source resistance adds to the 10.0k: keep R <= 100 R (1 % gain) or calibrate it out. AN1 = inductor current (INT-12, one constant); PVCELL-25 IEM LA 150-P 20.0 mV/A, 1.25 V CM, < 1 R source, linear +/-11k A -> 8.0k mV/A at the ADC. 0.0759 A/V/LSB; CMPS5 backup +/-91.5 A. The LA 150-P offset (up to 1.3 A) is re-zeroed by firmware at every idle (gates off); trip bands carry only the 1.17 A left after it. DAB-360 (C1/C2): AN1 5.009 mV/A (+/-1.58 V max), AN2 <= 2.79 V, AN3 <= 2.73 V single-ended, 49.9 R sources (+0.5 % gain). Only AN1 carries the open-input bias. Channel D: VMID follower.

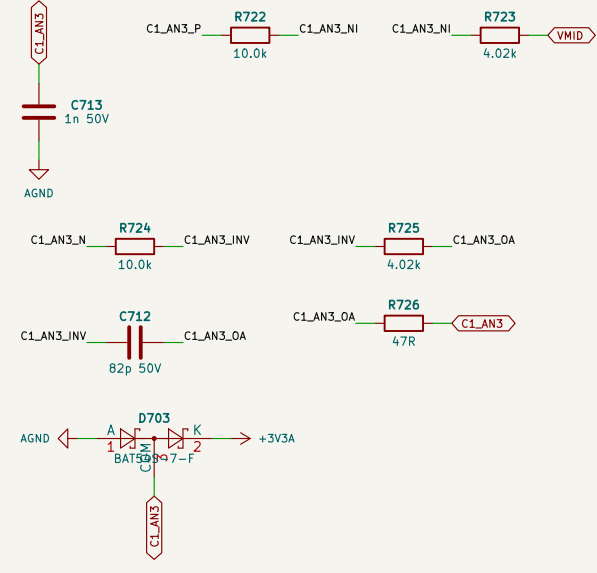
C1_AN1 -> ADCINA2 / CMPIN1P



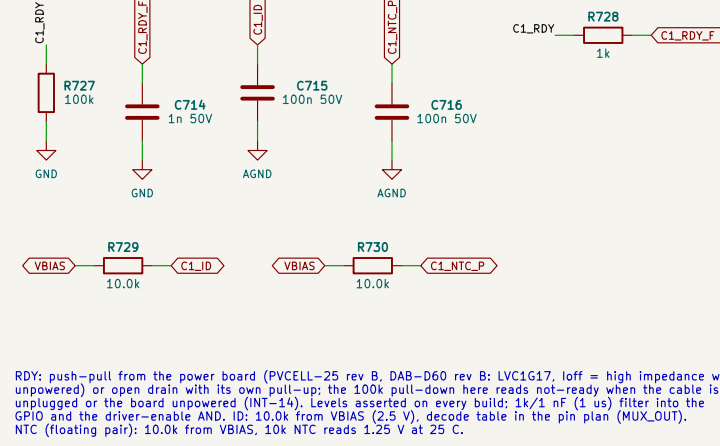
C1_AN2 -> ADCINB3



C1_AN3 -> ADCINC3



RDY, ID, NTC



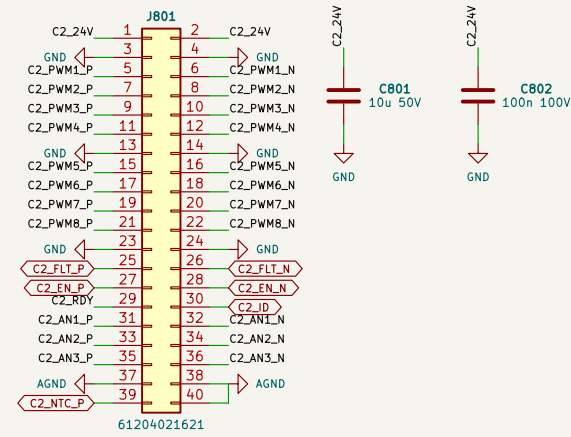
RDY: push-pull from the power board (PVCELL-25 rev B; DAB-D60 rev B; LVC1G17, loff = high impedance when unpowered) or open drain with its own pull-up; the 100k pull-down here reads not-ready when the cable is unplugged or the board unpowered (N1-14). Levels asserted on every build; 1k/2 nF (1 us) filter into the GPIO and the driver-enable AND; ID: 10.0k from VBIAS (2.5 V), decode table in the pin plan (MUX_OUT). NTC (floating pair): 10.0k from VBIAS, 10k NTC reads 1.25 V at 25 C.

Not bench-validated. Values marked calculated are not simulated.
Sheet 8
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

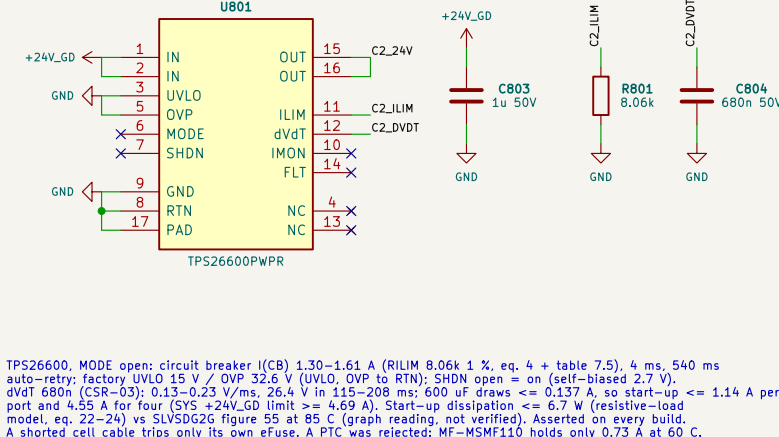
Sheet: /07 Cell port C1/
File: 07_cell1.kicad_sch
Title: CTRL-C2000 control card - Cell port C1

Size: A2 Date: 2026-10-04 Rev: E0
KiCad E.D.A. 10.0.6 Id: 8/15

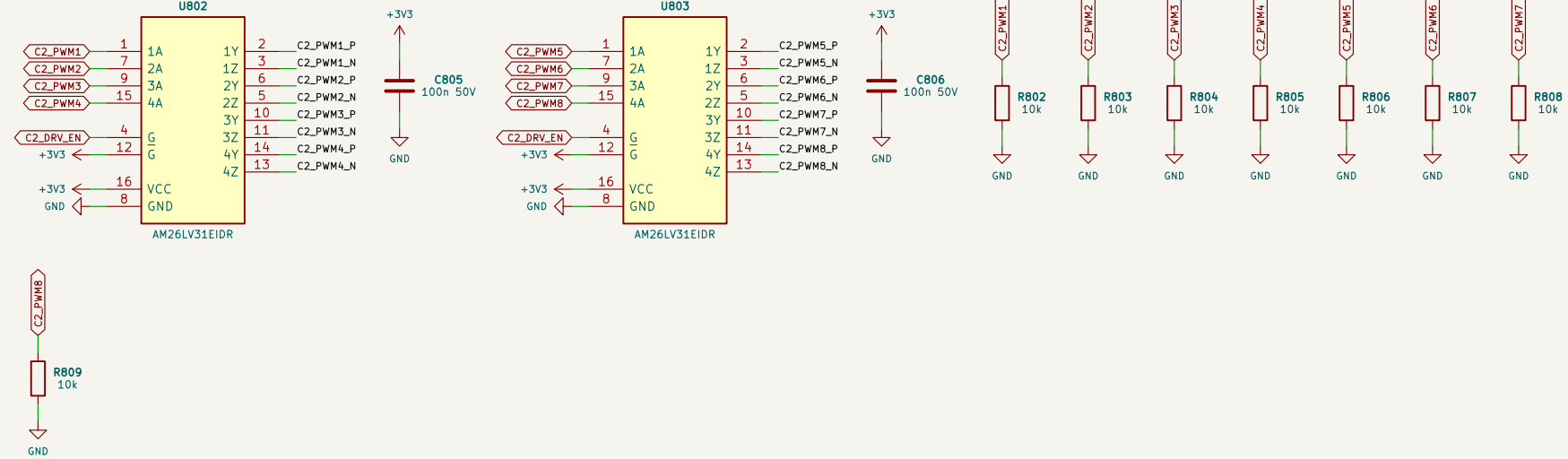
C2 CONNECTOR (CELL CONTRACT)



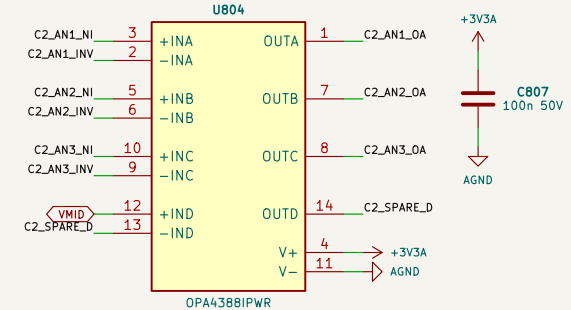
+24V_GD EFUSE FOR C2 (<= 1.0 A CONTINUOUS, <= 600 uF)



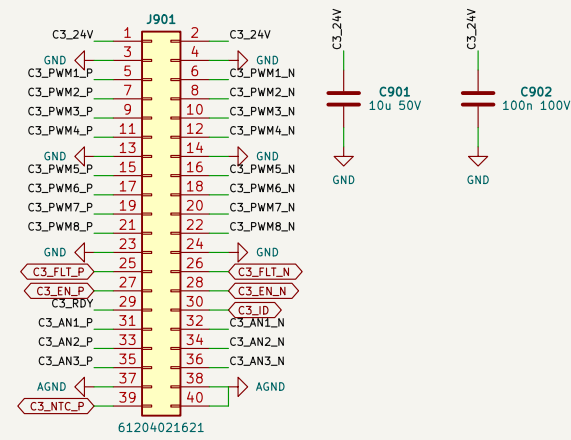
PWM LINE DRIVERS (EPWM5-8)



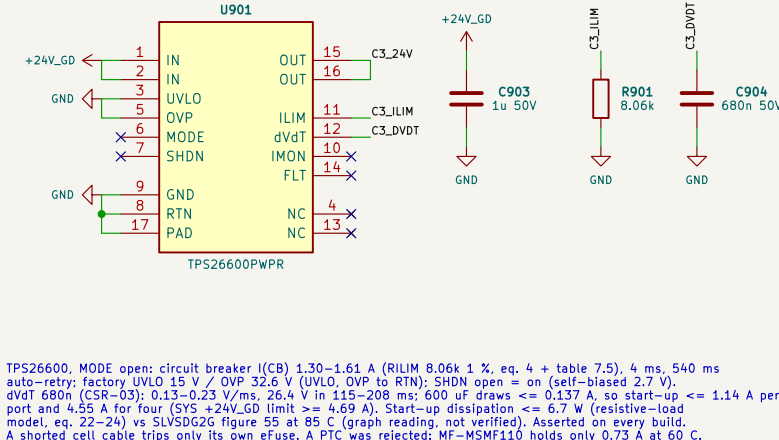
ANALOG FRONT ENDS AN1-AN3 (OPA4388), DUAL RANGE



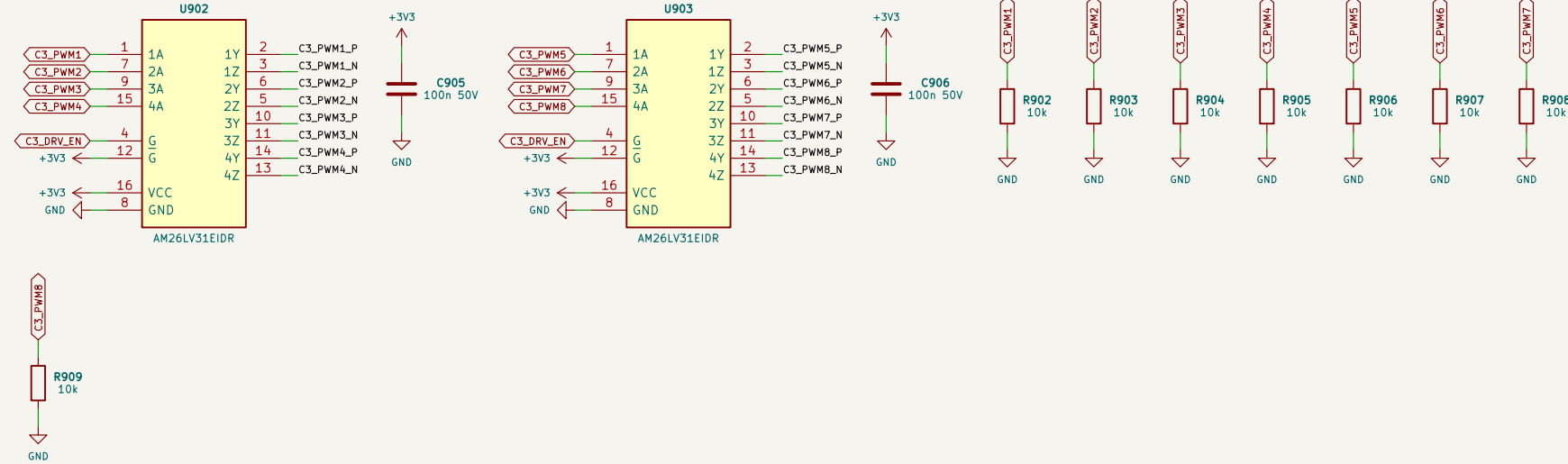
C3 CONNECTOR (CELL CONTRACT)



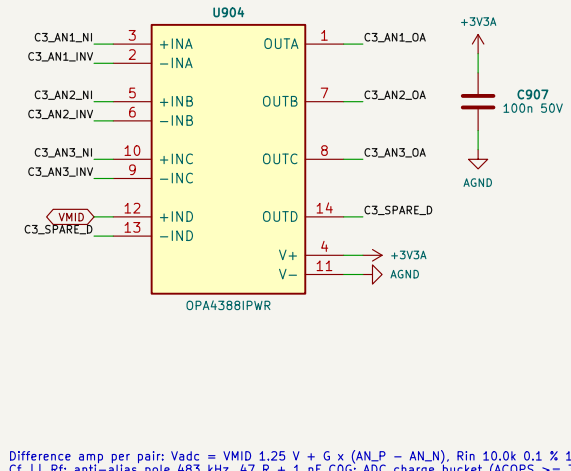
+24V_GD EFUSE FOR C3 (<= 1.0 A CONTINUOUS, <= 600 uF)



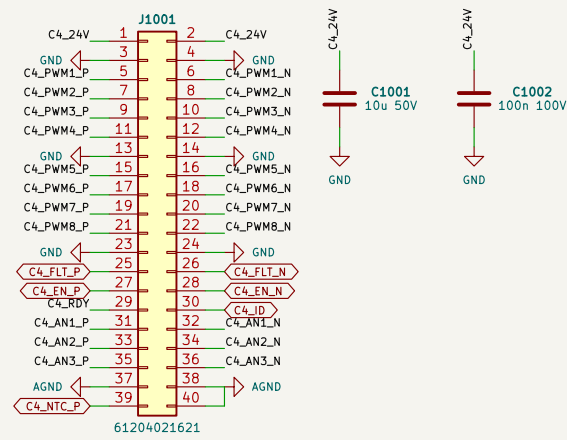
PWM LINE DRIVERS (EPWM9-12)



ANALOG FRONT ENDS AN1-AN3 (OPA4388), DUAL RANGE

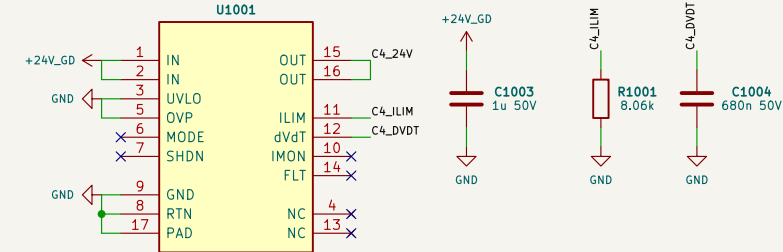


C4 CONNECTOR (CELL CONTRACT)



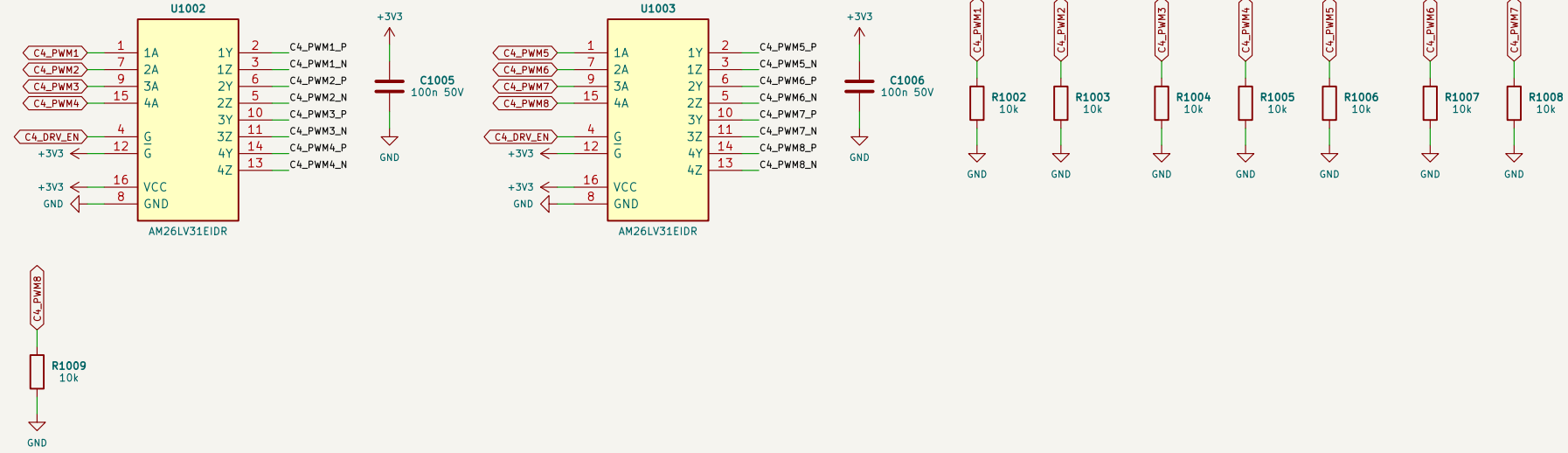
Pin map = Interfaces.CELL. Pins 1-2 carry +24V_GD only (gate-driver bias, channel B switched).
NTC_N returns to AGND here. FLV/EN gains go to the trip sheet.

+24V_GD EFUSE FOR C4 (<= 1.0 A CONTINUOUS, <= 600 uF)



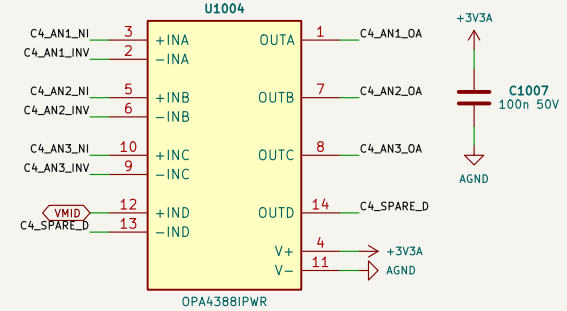
TPS26600, MODE open: circuit breaker I(CB) 1.30-1.61 A (RILIM 8.06k 1 %, eq. 4 + table 7.5), 4 ms, 540 ms auto-retry; factory UVLO 15 V / OVP 32.6 V (UVLO, OVP to RTN); SHDN open = on (self-biased 2.7 V).
dVdI 680k (CSR-D3): 0.13-0.23 V/ms, 26.4 V in 115-208 ms; 600 uF draws <= 0.137 A, so start-up <= 1.14 A per port and <55 A for four (SYS <24V_GD limit <= 4.69 A). Start-up dissipation <= 6.7 W (resistive-load model, eq. 22-24) vs SLVSDG2G figure 55 at 85 C (graph reading, not verified). Asserted on every build.
A shorted cell cable trips only its own eFuse. A PTC was rejected: MF-MSMF110 holds only 0.73 A at 60 C.

PWM LINE DRIVERS (EPWM13-16)



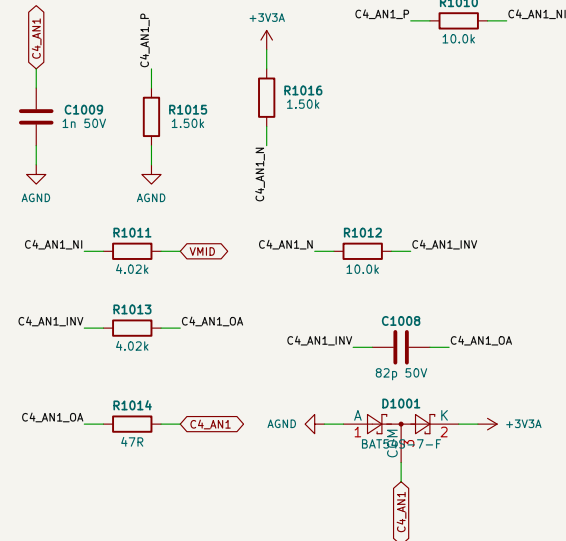
PWMs (1 = switch on) -> RS-422 pair, terminated on the power board. 10k pull-downs; ePWM balls are inputs without pull-up during reset/boost (table 6-8), so every gate command is 0 until firmware runs.
G = C4_DRV_EN, G- high; outputs high impedance unless CHB_OK, C4_EN_MCU and RDY are high (INT-15).
ePWM13-16 have no high-resolution PWM; buck-boost cells only, not a DAB bridge.

ANALOG FRONT ENDS AN1-AN3 (OPA4388), DUAL RANGE

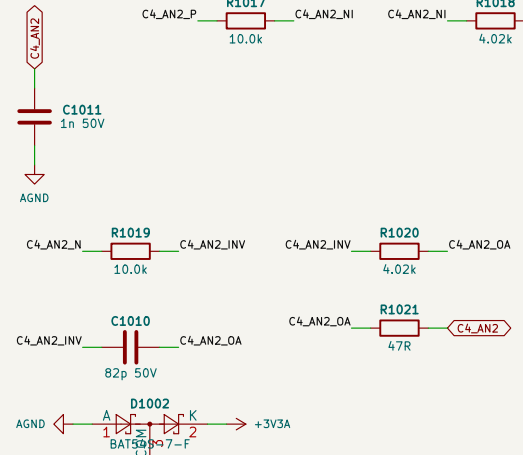


Difference amp per pair: $V_{\text{dcb}} = V_{\text{MID}} 1.25 \text{ V} + G \times (ANL_P - ANL_N)$, Rin 10.0k 0.1 % 10 ppm/K, $G = 4.02k/10.0k = 0.402$.
 C_f || RF: anti-alias pole 483 kHz, 47 R + 1 nF COG; ADC charge bucket (ACQPS >= 320 ns at 12 bit).
BAT5AS clamps the ADC node to 3V3A/AGND. All values calculated, not simulated.
Accepts on any AN pair differential on a 1.24-1.49 V common mode, legs 0.1-2.6 V (ADC span +/-3.1 V), or 0-3.0 V single-ended with $ANL_N = \text{AGND}$ at the source (temperatures) -> 1.25-2.46 V. Op-amp inputs 0.9-1.75 V (RRIO). Source resistance adds to the 10.0k; keep $R_s <= 100 \text{ R}$ (1 % gain) or calibrate it out.
AN1 = inductor current (INT-12, one constant); PVCELL-26 LEM LA 150-P 20.0 mV/A, 1.25 V CM, < 1 R source, linear +/-114 A -> 8.04 mV/A at the ADC. 0.0759 A/LSD; CMPS5 backup +/-91.5 A. The LA 150-P offset (up to 1.3 A) is re-zeroed by firmware at every idle (gates off); trip bands carry only the 1.17 A left after it.
DAB-D60 (C1/C2): AN1 5.005 mV/A (+/-1.58 V max), AN2 <= 2.79 V, AN3 <= 2.73 V single-ended, 49.9 R sources (+0.5 % gain). Only AN1 carries the open-input bias. Channel D: VMID follower.

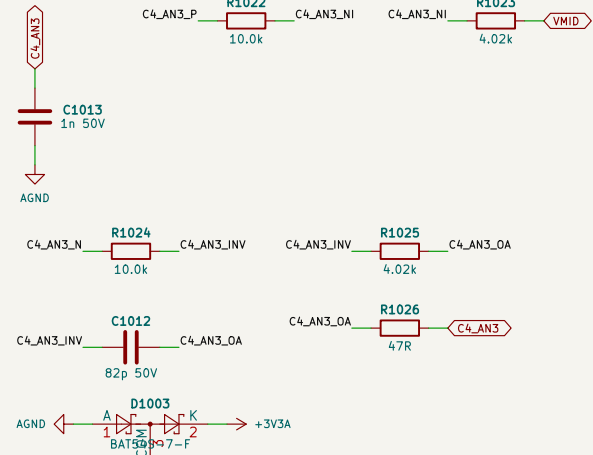
C4_AN1 -> ADCIND0 / CMPIN7P



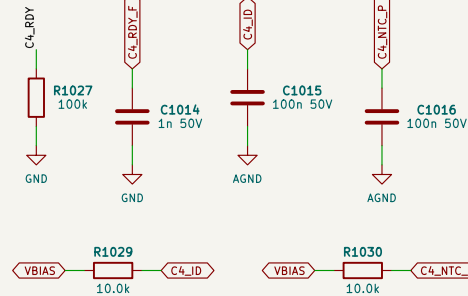
C4_AN2 -> ADCINA5



C4_AN3 -> ADCINB4



RDY, ID, NTC



RDY: push-pull from the power board (PVCELL-25 rev B; DAB-D60 rev B; LVC1G17, loff = high impedance when unpowered) or open drain with its own pull-up; the 100k pull-down here reads not-ready when the cable is unplugged or the board unpowered (INT-14). Levels asserted on every build: 1k/1 nF (1 us) filter into the GPIO and the driver-enable AND, ID: 10.0k from VBIAS (2.5 V), decode table in the pin plan (MUX_OUT).
NTC (floating pair): 10.0k from VBIAS, 10k NTC reads 1.25 V at 25 C.

Not bench-validated. Values marked calculated are not simulated.
Sheet 11
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

Sheet: /10 Cell port C4/
File: 10_cell4.kicad_sch
Title: CTRL-C2000 control card - Cell port C4

Size: A2 Date: 2026-10-04 Rev: E0
KiCad E.D.A. 10.0.6 Id: 11/15

[illegible]

U1102				U1103			
P_VA_INV	3	+INA	OUTA	P_IA_IN	3	+INA	OUTA
P_VA_INV	2	-INA		P_IA_INV	2	-INA	
P_VAX_IN	5	+INB	OUTB	P_IB_IN	5	+INB	OUTB
P_VAX_INV	4	-INB		P_IB_INV	6	-INB	
P_VBL_IN	10	+INC	OUTC	P_AUX1_IN	10	+INC	OUTC
P_VBL_INV	9	-INC		P_AUX1_INV	9	-INC	
P_VBK_IN	12	+IND	OUTD	P_AUX2_IN	12	+IND	OUTD
P_VBK_INV	13	-IND		P_AUX2_INV	13	-IND	
V+	4	+3V3A		V+	4	+3V3A	
V-	11	AGND		V-	11	AGND	

DPAA388IPWR DPAA388IPWR

P_JA -> ADCINA4 / CMPIN2P

[illegible]

The schematic diagram illustrates the power supply circuit for the P-VAX. It features a 50V input connected to a 100k resistor (R1110) and a 1.50k resistor (R1115). The circuit is powered by a +3V3A source and connected to AGND. Key components include resistors R1111 (5.76k), R1112 (10.0k), R1113 (5.76k), R1114 (47R), and capacitors C1109 (100nF) and C1109 (56pF). The P-VAX component is connected to the output of the circuit.

[illegible]

The schematic diagram illustrates the power supply section of the ADALM2000. It features a +3V3A input connected to a network of resistors (R1125, R1126, R1127, R1128, R1129, R1130, R1131), capacitors (C1114, C1113), and a diode (D1104). The circuit is grounded to AGND. Key components include a 50V capacitor C1114, a 56pF capacitor C1113, and a 47R resistor R1128. The output is labeled P_VBX.

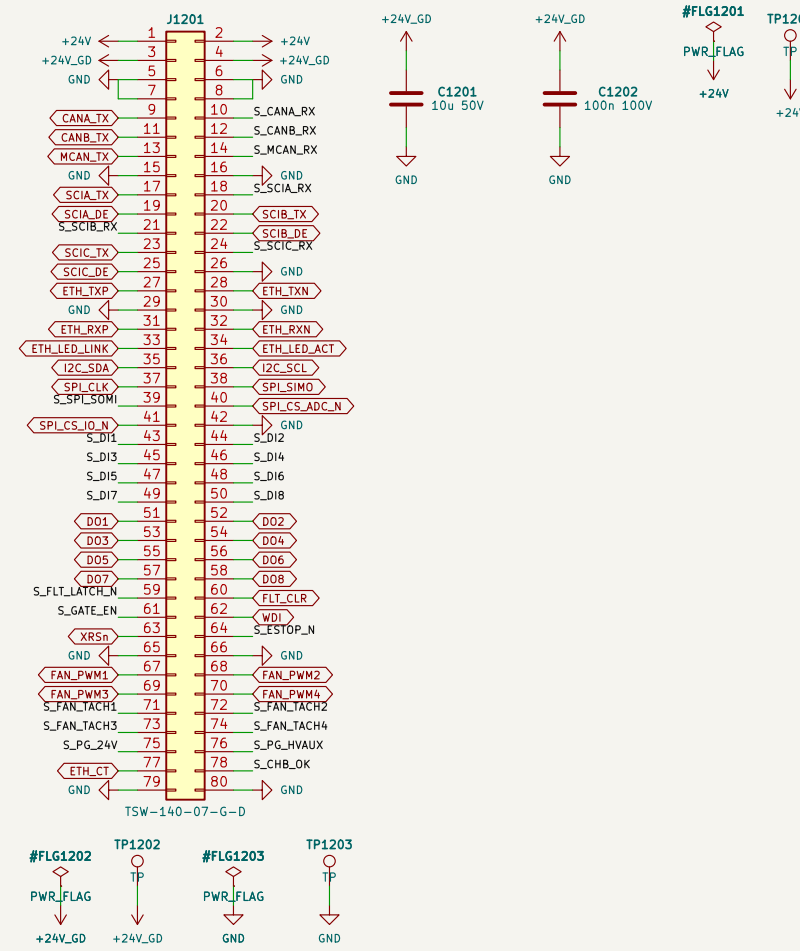
The schematic diagram illustrates the power supply section of the ADXL345 evaluation board. It features a 3V3A input source connected to a 10.0k resistor (R1136) in series. This series combination is connected to the VIN pin of the ADXL345. A 5.76k resistor (R1137) is connected in parallel from the VIN pin to ground (AGND). The output of the ADXL345 is connected to the VDD pin of the ADXL345, which is also connected to the VDD pin of the ADXL345. The output is connected to the VDD pin of the ADXL345, which is also connected to the VDD pin of the ADXL345.

[illegible]

The schematic diagram illustrates the P_AUX1 input signal path. It begins with a 50V source connected to a 10k resistor (R1120). This is followed by a 1.5k resistor (R1150) connected to AGND. The signal then passes through a 10k resistor (R1147) to the P_AUX1_NI input. The signal is then split into two paths: one through a 56pF capacitor (C1119) to the P_AUX1_INV input, and another through a 50V source (C1119) to the P_AUX1_OA input. The signal is also connected to the P_AUX1 block via a 47R resistor (R1149). The signal path is terminated at AGND and +3V3.

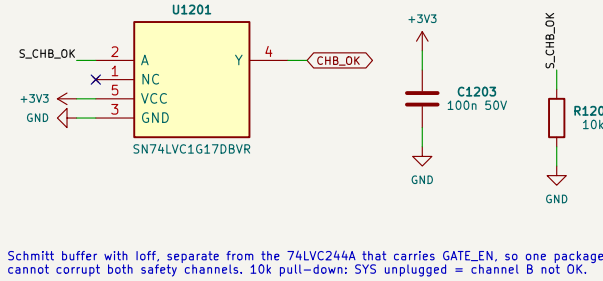
[illegible]

SYS CONNECTOR (SYS CONTRACT)



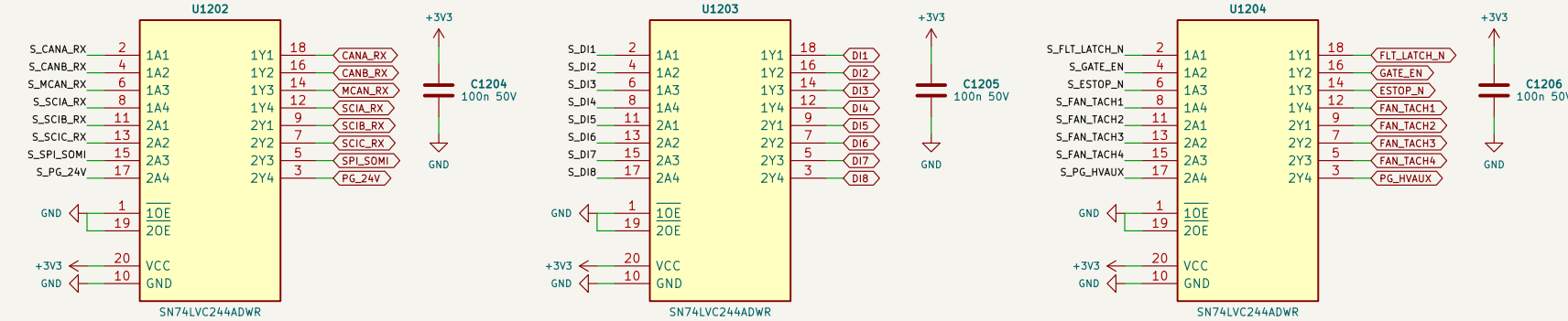
Pin map = interfaces.SYS: SYS_RST_N lands directly on XRSn (open drain both sides).
 +24V (pins 1-2): this card's regulators and the PORT connector. +24V_GD (pins 3-4): switched by safety channel. 18 on SYS: goes ONLY to the four cell eFuses (cell sheets). The two never meet here. ETH_CT (77): PHY analog supply to the magnetics centre taps (Ethernet sheet).
 Inputs from SYS arrive as `S_<name>` and reach the MCU as `<name>` through the buffers.

CHB_OK BUFFER (CHANNEL B, OWN PACKAGE)



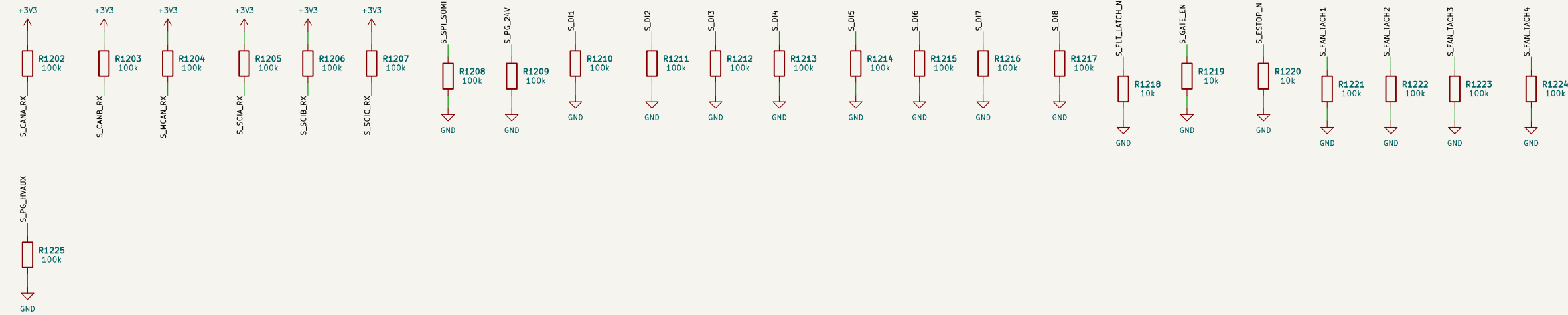
Schmitt buffer with 10k, separate from the 74LVC244A that carries GATE_EN, so one package cannot corrupt both safety channels. 10k pull-down: SYS unplugged = channel B not OK.

SYS INPUT BUFFERS (IOFF)



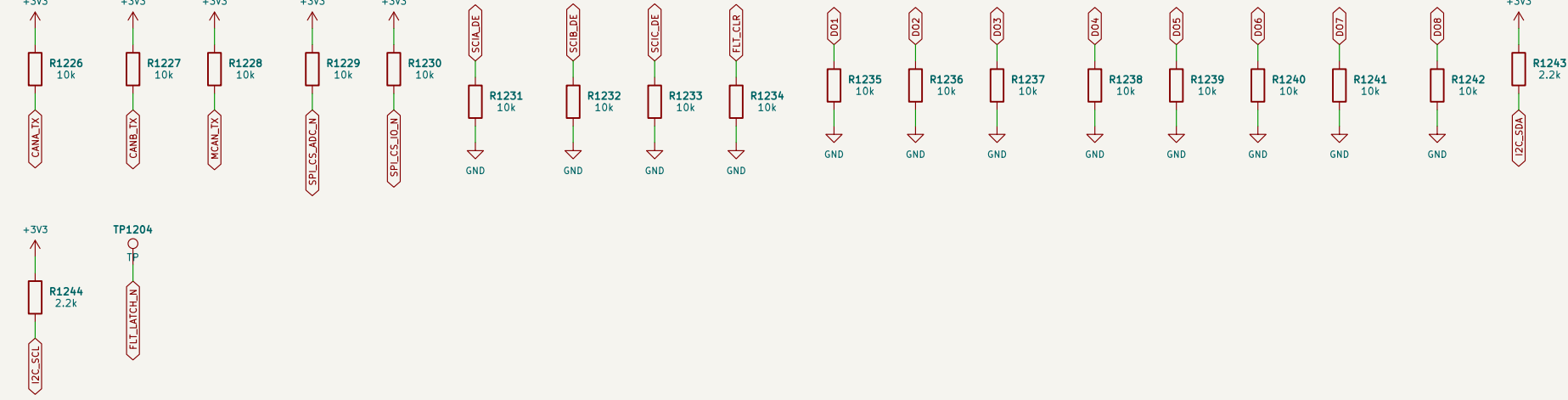
SYS makes its own 3.3 V and may drive these lines before this card's VDDIO is up; SPRSP14E 7.1 allows ± 20 mA total input-clamp current and 7.10.1.4.2 forbids driving pins early. 74LVC244A inputs and outputs are high impedance unpowered (loff) and its outputs never exceed VDDIO. tpd -5 ns.

INPUTS WITH SYS UNPLUGGED



FLT_LATCH_N, GATE_EN, ESTOP_N: 10k to GND, so an unplugged SYS reads as tripped and the gates stay off.
Other inputs: 100k to their idle level (CAN/SCI receive high, the rest low).

OUTPUTS WHILE THE MCU IS IN RESET



GPIOs float in reset: CAN TX recessive, RS-485 DE off, DO and FLT_CLR low, SPI chip selects high
I2C: 2.2k pull-ups here (bus master side); SYS should not add a second pair.

Not bench-validated. Values marked calculated are not simulated.
Sheet 13
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

Sheet: /12 SYS-I0-AUX connector/
File: 12_sys.kicad_sch

Title: CTRL-C2000 control card - SYS-IO-AUX connector

Size: A2

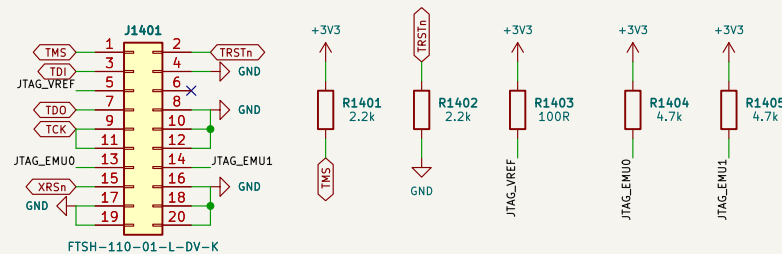
Date: 2026-10-04

Rev: E0

1.6

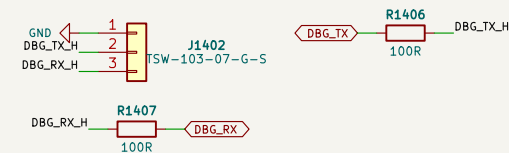
Id: 13/15

JTAG CTI-20 (SPRU655I TABLE 13)



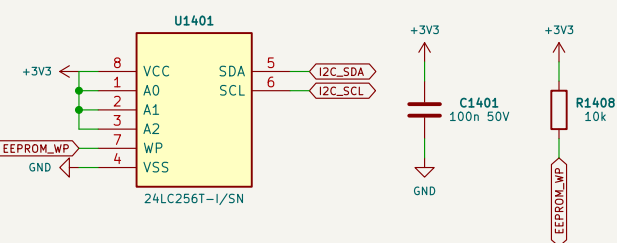
TMS 2.2k pull-up and TRSTn 2.2k pull-down (SPRSP14E table 6-1); TDIS to GND; VTRef via 100 R; RTCK looped to TCK; EMU0/1 4.7k pull-ups (no EMU pins on F2838x); RESET (open drain) on XRSn; EMU2-EMU4 (17-19) grounded as SPRSP14E 7.10.7 requires (CSR-07).

DEBUG UART (CM-UART-A, 3.3 V TTL)



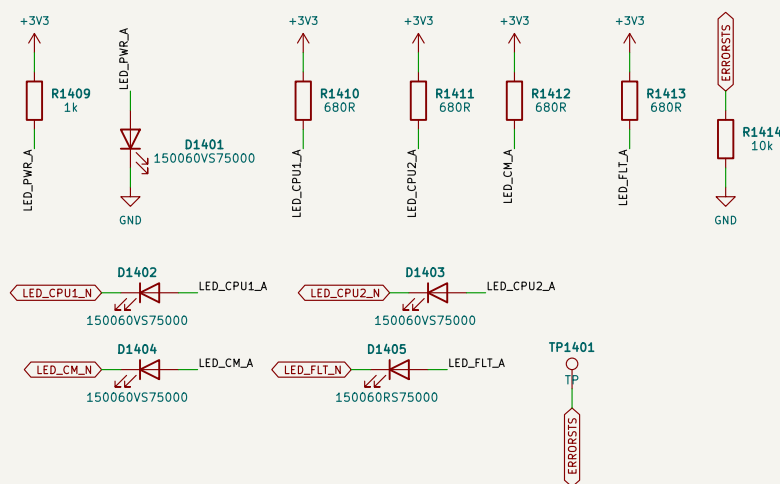
Pin 1 GND, 2 TX (from MCU), 3 RX (to MCU); 100 R series each way.

CALIBRATION / SERIAL-NUMBER EEPROM



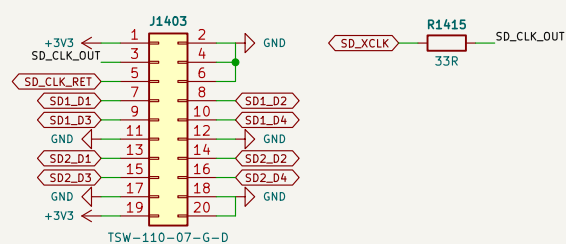
Address 1010111 (A2-A0 high = 0x57) so it cannot collide with a SYS EEPROM at the default 0x50.
WP pulled high: write-protected unless firmware drives EEPROM_WP low.

STATUS LEDS AND ERRORSTS



GPIO sinks about 2 mA (680 R). ERRORSTS needs an external pull-down.

SDFM EXPANSION HEADER (NO MATING CONTRACT YET)



8 modulator data inputs (SD1/SD2 D1–D4) and one clock: XCLKOUT out, returned clock to all eight SDx-Cy inputs. 3.3 V CMOS; for an adjacent sensing board only.